Midterm

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GRA6551 Quantitative Risk And Asset Management

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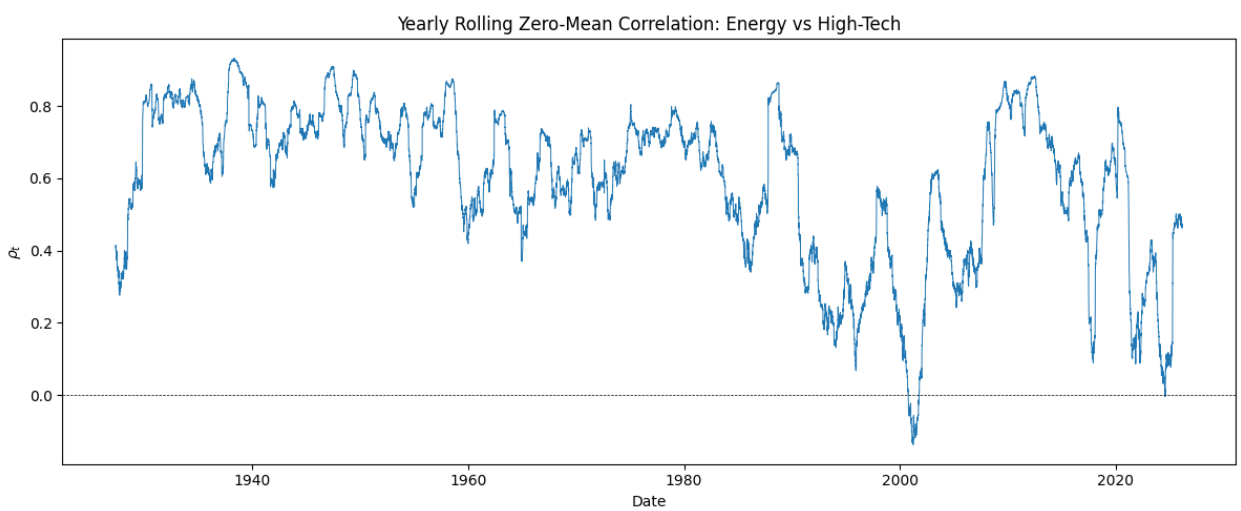
1 Rolling zero-mean correlations of returns

For the full sample, compute yearly rolling zero-mean correlations of returns between the energy and high-tech sector. This is done as follows: at each date t (excluding the first 252 days), compute the correlation between the daily returns of the energy and high-tech sector, but using a zero-mean rather than the sample mean. This correlation is then:

$$\rho_t = \frac{\frac{1}{252}\sum_{j=1}^{252} R_{E,t-j} R_{HT,t-j}}{\sqrt{\frac{1}{252}\sum_{j=1}^{252} R_{E,t-j}^2}\sqrt{\frac{1}{252}\sum_{j=1}^{252} R_{HT,t-j}^2}}$$

1. Rolling correlation plot

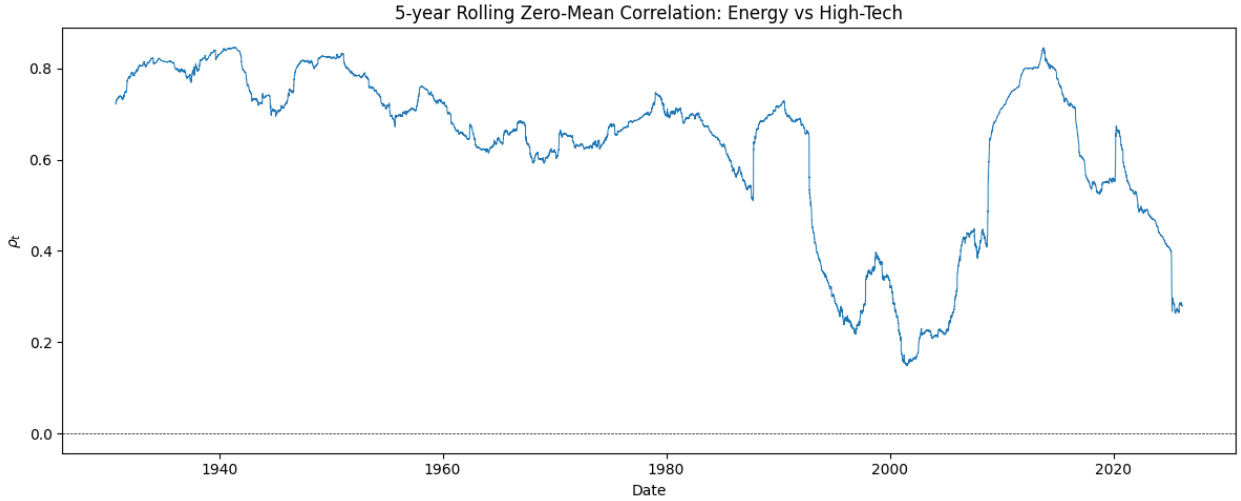


2. Compute its mean, median, q 0.05 and q 0.95

   This was done using simple `np` functions:

mean	**median**	**q 0.05**	**q 0.95**
0.601	0.656	0.182	0.861

3. Repeat 1-2 with a 5-year rolling correlation



mean	**median**	**q 0.05**	**q 0.95**
0.634	0.685	0.236	0.824

4. Briefly comment on the results